

Small High-Efficiency Broadband Rectifier Based on Compact Coupled structure

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Abstract — In this paper, a high-efficiency broadband rectifier is proposed, which relies on a small coupled matching structure to ensure the compactness and high efficiency of the rectifier. The proposed rectifier incorporates a series-coupled matching network and cascaded band-stop structures, with the wideband response mechanisms analyzed through theoretical modeling. Highly compact rectifiers are designed, fabricated, and measured to validate the proposed approach. The broadband rectifier, operating between 1.8 and 3.0 GHz with a relative bandwidth of 50%, achieves above 70% rectification efficiency and a peak efficiency of 79%. Moreover, the physical dimension of the rectifier is $15.8 \times 18.9 \text{ mm}^2$, demonstrating its compactness. It is highly compact with high rectification efficiency, making it suitable for wireless power transmission systems and energy harvesting applications.

Keywords — broadband, coupled structure, energy harvesting, rectifier, wireless power transfer.

I. INTRODUCTION

Microwave Wireless Power Transfer (MWPT) and Energy Harvesting (EH) technologies have attracted considerable research interest due to their vast potential applications across various sectors[1][2]. These technologies improve device convenience and safety while minimizing dependence on conventional cables and batteries, thereby supporting environmental sustainability. A critical component of these systems is the rectifier, which plays a key role in efficiently converting received electromagnetic energy. Ensuring high efficiency and stability across a wide frequency range remains a significant design challenge for rectifiers in MWPT and EH applications. Moreover, the compactness of the rectifier is a critical design factor, as miniaturization enhances system integration and meets the growing demand for high-density, efficient power conversion in modern wireless energy transmission applications.

Expanding bandwidth can be achieved through broadband matching [3][4] and impedance compression techniques. Non-uniform transmission lines have been employed to optimize impedance matching, enabling high rectification efficiency over a wide frequency range [5]. Additionally, impedance compression minimizes input impedance variations, leading to a more compact and efficient rectifier design [7][8][9]. Wideband conjugate matching between the antenna and diode improves the integration and overall performance of the system [10][11].

This paper presents an innovative design approach for compact wideband rectifiers, emphasizing both structural simplicity and miniaturization. The proposed design integrates

a cascaded coupled matching network (CMS) with a series band-stop structure to achieve efficient broadband impedance matching. The impedance characteristics of these structures are analytically derived to facilitate optimized design and performance evaluation.

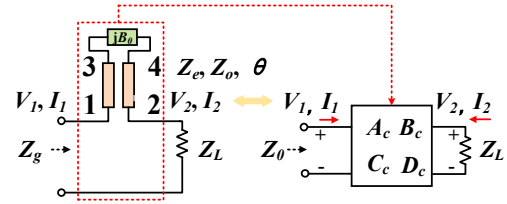


Fig. 1. Schematic diagram of the proposed coupled matching structure and its equivalent two-port network.

II. PRINCIPLE AND DESIGN METHOD

A. Principle of the Coupled Matching Structure

The schematic of the proposed coupled matching structure is shown in Fig. 1. Unlike traditional coupled-line networks, ports 3 and 4 are series-connected through a reactive element, B_0 . The coupled line has an electrical length θ , with characteristic impedances Z_e and Z_o for the even and odd modes, respectively. The impedance between ports 1 and 2 is modeled as a series element using an ABCD matrix, which facilitates impedance matching between the system impedance (Z_g) and the load impedance (Z_L).

[12] introduced an innovative circuit analysis approach that combines the port voltage-current method with even-odd mode decomposition for the analysis of coupled-line structures. This method effectively reduces the four-port coupled-line network to an equivalent two-port network. Two ABCD matrices functioning in even and odd modes are obtained, which are respectively denoted as [13]

$$\begin{bmatrix} A_e & B_e \\ C_e & D_e \end{bmatrix} = \begin{bmatrix} \cos \theta & jZ_e \sin \theta \\ j \sin \theta / Z_e & \cos \theta \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} A_o & B_o \\ C_o & D_o \end{bmatrix} = \begin{bmatrix} \cos \theta - 2B_0 Z_o \sin \theta & jZ_o \sin \theta \\ j(\sin \theta / Z_o + 2B_0 \cos \theta) & \cos \theta \end{bmatrix} \quad (2)$$

The resulting ABCD matrix for this network, which represents the CMS, is expressed as follows

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \quad (3)$$

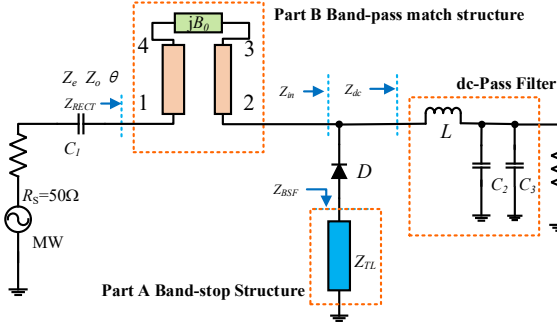


Fig. 2. Proposed wideband rectifier I with the CMS.

where

$$\begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} = \begin{bmatrix} \frac{A_e C_o + A_o C_e}{A_e C_o - A_o C_e} & \frac{2A_e A_o}{A_e C_o - A_o C_e} \\ \frac{2C_e C_o}{A_e C_o - A_o C_e} & \frac{A_e C_o + A_o C_e}{A_e C_o - A_o C_e} \end{bmatrix} \quad (4)$$

The two-port network illustrated in Fig. 1 efficiently connects the system input to the load. A low voltage reflection coefficient helps minimize return losses, thereby improving impedance matching. Therefore, it is essential to ensure that the CMS provides effective impedance transformation. By analyzing the $ABCD$ matrix, the voltage reflection coefficient can be expressed as

$$S_{11} = \frac{A_c Z_L + B_c - C_c Z_g Z_L - D_c Z_g}{A_c Z_L + B_c + C_c Z_g Z_L + D_c Z_g} \quad (5)$$

where $Z_L = R_L + jX_L$ is the load impedance, and R_L is greater than zero.

As an ideal impedance match with a CMS, it is required to satisfy the condition $S_{11} = 0$. Thus, when $Z_g = R_S$ is the source resistance, $R_L \neq R_S$, (5) is simplified as

$$Z_e = \left(-M_1 + \sqrt{M_1^2 + M_2} \right) \tan \theta \quad (6)$$

$$B_0 = \frac{M_2 \tan^2 \theta + Z_e Z_o}{2(Z_e Z_o - M_2) Z_o \tan \theta} \quad (7)$$

where

$$M_1 = \frac{R_S X_L}{R_L - R_S} \quad (8)$$

$$M_2 = \frac{R_S X_L^2}{R_L - R_S} + R_S R_L \quad (9)$$

Equations (6) to (9) demonstrate that Z_o and θ are two independent variables, unaffected by the load impedance (Z_L) or source impedance (R_S). These parameters can be freely adjusted, providing flexibility in the design of the CMS for broadband rectifier matching. Through this analysis, the CMS design effectively converts any complex impedance to match the source impedance. As a result, the rectifiers utilize this configuration as an impedance-matching network, thereby improving both conversion efficiency and operational bandwidth.

B. Design of the rectifier

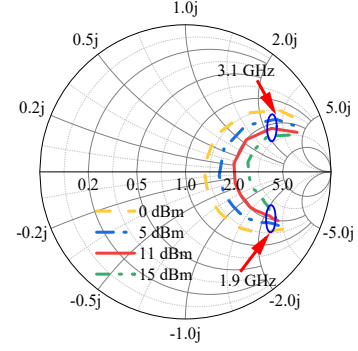


Fig. 3. Z_{in} over 1.9-3.1 GHz at various input power levels.

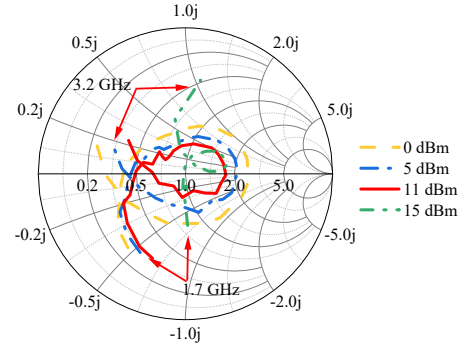


Fig. 4. Z_{RECT} over 1.7-3.2 GHz at different input power levels.

Fig. 2 illustrates the schematic of the proposed wideband rectifier. The rectifier consists of key components, including the isolation capacitor C_1 , a DC-pass filter, and a Schottky diode D . Part A is a band-stop structure formed by a series $\lambda_g/8$ short-ended transmission line, while Part B represents the CMS. With minimal distributed elements, the rectifier is highly compact.

The input impedance of the $\lambda_g/8$ short-ended transmission line is

$$Z_{BSF} = jZ_{TL} \tan\left(\frac{\pi f}{4 f_0}\right) = \begin{cases} jZ_{TL} \tan\left(\frac{\pi f}{4 f_0}\right) & 0 < f < f_0 \\ jZ_{TL} & f = f_0 \\ jZ_{TL} \tan\left(\frac{\pi f}{4 f_0}\right) & f_0 < f < 2f_0 \\ \infty & f = 2f_0 \end{cases} \quad (10)$$

where Z_{TL} is the characteristic impedance of the transmission line. It can effectively adjust diode impedance. It enables impedance matching at f_0 , while Z_{BSF} becomes infinite at $2f_0$, effectively recycling the second harmonic at $2f_0$ during rectification. This mechanism enhances rectification efficiency. Z_{in} becomes the impedance of the diode in series with the band-stop structure as

$$Z_{in} = Z_D + Z_{BSF} \quad (11)$$

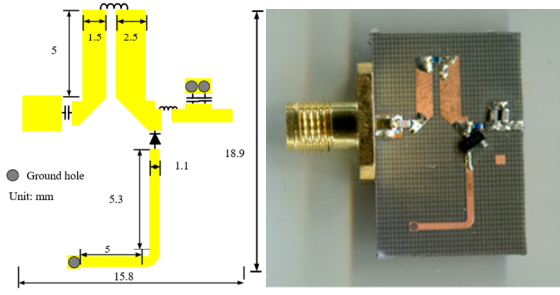


Fig. 5. Layout and photograph of the rectifier with parameter details.

Table 1. Spice parameters of HSMS286 diode

B_v	C_{j0}	I_s	R_s	V_{bi}
7.0 V	0.18 pF	0.05 μ A	6.0 Ω	0.25 V

Thus, Z_{in} is capacitive when $f < f_0$ and inductive when $f_0 < f < 2f_0$. The imaginary part of the impedance of the diode series with the band-stop structure at an input power level is

$$\text{Im}(Z_{in})|_{f=f_0} = \text{Im}(Z_D + Z_{BSF})|_{f=f_0} = 0 \quad (12)$$

Due to the analysis of the band-stop structure, Z_{in} is approximately located along a constant resistance circle on the Smith chart in the operating frequency band, as shown in Fig. 3. The load impedance Z_L is set to $Z_L = R_L$, $Z_{f1} = R_L - jX_{L1}$ and $Z_{f2} = R_L + jX_{L1}$ at center frequency f_0 , start frequency f_1 and the stop frequency f_2 , respectively.

For optimal performance, the rectifier's input impedance must match the characteristic impedance of the system, i.e., $Z_0 = 50 \Omega$. Therefore, the source impedance is set to $Z_0 = R_s = 50 \Omega$. Substituting these values into equations (8) and (9) yields

$$M_1 = \frac{Z_0 X_L}{R_L - Z_0} \quad (13)$$

$$M_2 = \frac{Z_0 X_L^2}{R_L - Z_0} + Z_0 R_L \quad (14)$$

Since the even-mode impedance Z_e is quite stable with frequency variation, we can establish equations to determine θ and complete the impedance matching at frequencies f_1 and f_2 . Because the load impedance Z_L is almost conjugate to each other at frequencies f_1 and f_2 , M_2 is identical at f_1 and f_2 , $M_2(f_1) = M_2(f_2)$. M_1 is inversed at f_1 and f_2 , $M_1(f_1) = -M_1(f_2)$. We take the electrical length θ_0 of the center frequency as the electrical length of the CMS. We derive a θ_0 that allows conjugate impedance matching at the start and stop frequencies.

Through analysis, the susceptive element is an inductor L_0 . Under specific inductance, B_0 changes with the frequency, $B_0(f_0)$ and $B_0(f_2)$ are the susceptance corresponding to center frequency f_0 and cutoff frequency f_2 . By leveraging the inductance L_0 to solve $B_0(f_0)$ and $B_0(f_2)$, the characteristic impedance Z_0 is derived, ensuring simultaneous impedance matching at the frequencies f_0 and f_2 . At this point, CMS achieves a match at f_0 , f_1 and f_2 .

Achieving impedance matching across the entire operating frequency band is challenging. However, it is reasonable to match the impedance at the three key frequency points: the

start f_1 , center f_0 , and stop f_2 frequencies. All parameters were subsequently optimized through simulation software to refine the performance and achieve enhanced results. As illustrated in Fig. 4, the rectifier's input impedance is approximately 50Ω , which ensures proper impedance matching for the rectifier.

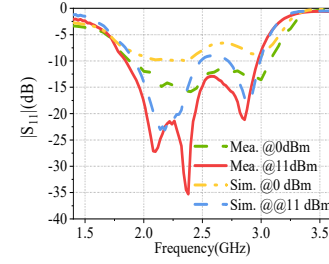


Fig. 6. S-parameter curves of the broadband rectifier.

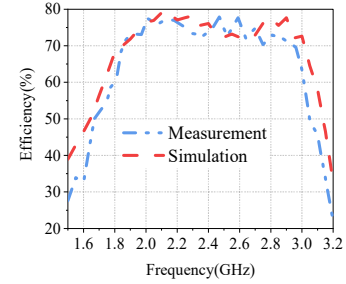


Fig. 7. Efficiency curves of broadband rectifier I at 11 dBm.

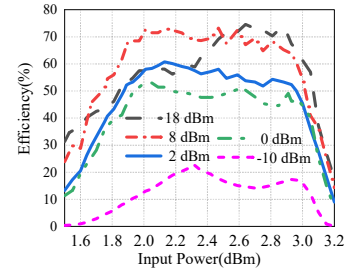


Fig. 8. Efficiency curves of broadband rectifier I at different powers.

III. RECTIFIER DESIGN AND MEASUREMENT RESULTS

To verify the feasibility of the design theory, a rectifier was fabricated based on the proposed design with an HSMS286 Schottky diode, whose spice parameters are shown in Table 1. The layout and physical prototype of the broadband rectifier is shown in Fig. 5, with F4B-2 substrate (dielectric constant of 2.65, and loss tangent of 0.0012). The specific design parameters are detailed in Fig. 5 as well, and the inductance of the susceptible element is 2.7 nH.

The simulated and measurement results of S-parameters are shown in Fig. 6. It can be observed that the simulation and measurement results align closely. Under RF power levels of 0 dBm and 11 dBm, $|S_{11}|$ remains below -10 dB across the frequency range of 1.8 GHz to 3 GHz, indicating good impedance matching.

Then, the power conversion efficiency (PCE) of rectifier was measured. The simulation and measurement results of rectification efficiency at 11 dBm are shown in Fig. 7, with a DC load of 500 Ω . The simulated efficiency closely matches

the measured results. The rectification efficiency exceeds 70% across the frequency band of 1.8 GHz to 3 GHz, achieving a fractional bandwidth of 50%. The rectifying efficiency stays above 50% from 1.88 to 2.98 GHz while the input power varies from 2 to 18 dBm, as shown in Fig. 8.

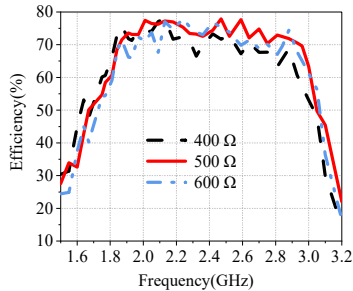


Fig. 9. RF-dc conversion efficiency with various dc loads at the input power of 11 dBm.

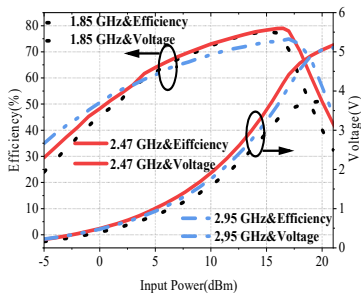


Fig. 10. Conversion efficiency and output dc voltage from -5 dBm to 22 dBm at 1.85 GHz, 2.47 GHz, and 2.95 GHz.

Table 2. Comparison between Broadband Rectifiers

Ref	PCE>70% (GHz)	BW (%)	Power (dBm)	PCE > 40% at 0 dBm (GHz)	BW (%)	Size (λ_g^2)
[3]	2.08-2.58	21.5	15.5	2.03-2.75	30	1.56×0.84
[4]	0.57-0.90	44.9	12.8	N. A	N. A	0.31×0.40
[5]	2.0-3.05	41.5	10	1.9-3.1	48	0.58×0.56
[8]	2.1-3.3	44.4	14	N. A	N. A	0.45×0.26
This Work	1.8-3	50	11	1.85-3.05	49	0.20×0.25

Fig. 9 shows the conversion efficiency as a function of various load resistances, with an input power of 11 dBm. The optimal DC load resistance is identified as 500 Ω .

Fig. 10 illustrates the impact of input power on both rectification efficiency and output voltage at 1.85 GHz, 2.47 GHz, and 2.95 GHz, corresponding to the rectifier's start, center, and stop frequencies. The results indicate that the highest measured efficiency, 79.0%, occurs with an input power of 16 dBm at 2.47 GHz. Moreover, the rectifier maintains an efficiency greater than 50% across the input power range of 0 to 20 dBm.

Table 2 compares the rectifier designed in this work with some previous designs. The rectifier designed using CMS exhibits superior bandwidth performance while maintaining

significantly smaller physical dimensions compared to similar rectifier types.

IV. CONCLUSION

This paper presents a novel design for small broadband rectifiers, incorporating a band-stop structure to optimize diode impedance and enhance rectification efficiency. A coupled matching network is employed to achieve effective impedance matching to the system across a broad bandwidth. The underlying mechanism is thoroughly explored, and comprehensive design methodologies are outlined. The design methodology enables the efficient development of broadband, compact rectifiers, enhancing their integration with antenna systems by simplifying the design process.

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